

LINEAR ALGEBRA

A small field guide

Linear algebra begins with a simple idea: complicated relationships can be described by combining a few familiar directions. A vector is a way to keep track of those directions and their sizes.

This short, original guide is included with Rhine Archive so you can explore the reader without importing a book. Try selecting a sentence, adding a bookmark, or opening two different pages side by side.

Read slowly. Draw the picture. Then test your understanding with a small example.

READING MAP

Vectors / Linear combinations / Matrices / Systems
Geometry / Exercises / Worked solutions

01 / THE BASIC OBJECTS

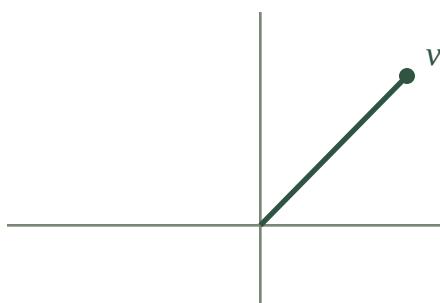
Vectors and direction

A vector in the plane has two components. We write it as $v = (x, y)$. The first component tells us how far to travel horizontally, and the second tells us how far to travel vertically.

For example, the vector $v = (3, 2)$ moves three units to the right and two units upward. Changing the starting point does not change the vector: direction and displacement are what matter.

Vector addition combines displacements. Starting at the origin, first follow u and then follow v . The endpoint is the same as following the single vector $u + v$.

Think in pictures.

**TRY IT YOURSELF**

Let $u = (1, 2)$ and $v = (3, -1)$.

Calculate $u + v$, then draw all three vectors.

02 / BUILDING WITH VECTORS

Linear combinations

Multiplying a vector by a scalar changes its length. A negative scalar also reverses its direction. For instance, $2(1, 3) = (2, 6)$, while $-1(1, 3) = (-1, -3)$.

A linear combination is a sum of scaled vectors. Given vectors u and v , an expression of the form $a u + b v$ is a linear combination. The scalars a and b tell us how much of each direction to use.

The span of a collection of vectors is the set of all their linear combinations. Two nonparallel vectors in the plane span the whole plane. Parallel vectors span only one line.

A WORKED EXAMPLE

$$u = (1, 0), \quad v = (1, 2)$$

$$3u + 2v = 3(1, 0) + 2(1, 2) = (5, 4)$$

03 / FROM NUMBERS TO MOTION

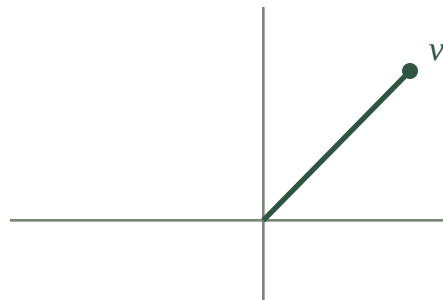
Matrices as transformations

A matrix can describe a transformation of space. Its columns tell us where the coordinate directions move. Once we know those two images, linearity determines where every other vector will go.

The identity matrix leaves every vector unchanged. A diagonal matrix scales the coordinate directions independently. A shear keeps one axis fixed while sliding points parallel to it.

Matrix multiplication represents composition. In the product AB , the transformation B acts first, followed by A . The order matters: in general, AB and BA are different.

Think in pictures.

**FOLLOW THE COLUMNS**

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad \begin{array}{l} A(1, 0) = (2, 0) \\ A(0, 1) = (1, 1) \end{array}$$

04 / LOOKING FOR AN INTERSECTION

Solving a linear system

A system of two linear equations can be viewed as two lines. A solution is a point lying on both lines. The lines may intersect once, be parallel and distinct, or coincide.

Elimination simplifies a system without changing its solution set. We can exchange equations, multiply an equation by a nonzero number, or add a multiple of one equation to another.

A useful habit is to check a computed solution in the original equations. Arithmetic errors are easier to detect with a direct substitution than with another pass through the same calculations.

ELIMINATION IN TWO STEPS

$$x + y = 5 \quad \text{and} \quad 2x - y = 1$$

Add the equations: $3x = 6$, so $x = 2$.

Substitute back: $y = 3$.

05 / MEASURING ALIGNMENT

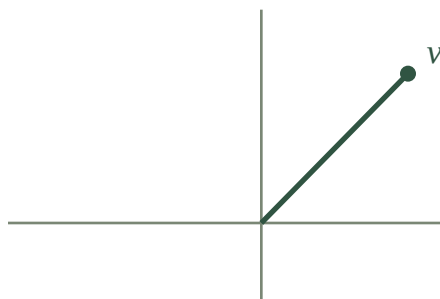
The geometry of a dot product

The dot product of $u = (u_1, u_2)$ and $v = (v_1, v_2)$ is $u_1 v_1 + u_2 v_2$. It turns a pair of vectors into a single number that measures how strongly their directions align.

For nonzero vectors, a positive dot product means the angle between them is less than a right angle. A negative value means the angle is greater than a right angle. A zero dot product means the vectors are perpendicular.

The squared length of a vector equals its dot product with itself. This gives a direct connection between algebraic computation and geometric distance.

Think in pictures.

**QUICK CHECK**

$$u = (2, 1), \quad v = (-1, 2)$$

$$u \text{ dot } v = 2(-1) + 1(2) = 0$$

These two vectors are perpendicular.

06 / MAKE THE IDEAS YOUR OWN

Exercises

1. Let $u = (2, 3)$ and $v = (-1, 4)$. Find $u + v$ and $2u - v$. Sketch the result of the addition using arrows.
2. Write the vector $(5, 4)$ as a linear combination of $a = (1, 0)$ and $b = (1, 2)$. Explain why the coefficients are unique.
3. Solve the system $x + y = 5$ and $2x - y = 1$. Verify your solution by substitution.
4. Find the dot product of $(2, 1)$ and $(-1, 2)$. What does the result tell you about their directions?
5. A transformation sends $(1, 0)$ to $(2, 0)$ and $(0, 1)$ to $(1, 1)$. Where does it send $(3, 2)$?

USE THE READER

Keep this page in the main pane. Open page 8 in the comparison pane to review the worked solutions.

07 / CHECK AND REFLECT

Worked solutions

1. Add components to obtain $u + v = (1, 7)$. Scaling and subtracting gives $2u - v = (4, 6) - (-1, 4) = (5, 2)$.
2. The second component gives $2b = 4$, so $b = 2$. The first component then gives $a + b = 5$, so $a = 3$. The two starting vectors are not parallel, which makes the coefficients unique.
3. Add the equations to get $3x = 6$. Thus $x = 2$ and $y = 3$. Check: $2 + 3 = 5$ and $2(2) - 3 = 1$.
4. The dot product is $2(-1) + 1(2) = 0$. The vectors are perpendicular.
5. By linearity, the image is $3(2, 0) + 2(1, 1) = (8, 2)$.

A LAST THOUGHT

Before moving on, explain one solution in your own words. A short note is often more useful than a highlight.